

Postura - Step 5: Workstation Questions

Developer instructions – March 2026

1 WHAT THIS STEP DOES

Add one new screen and one new dropdown to an existing screen. These collect 7 inputs that the photo cannot detect. Without these inputs, the ROSA score is incomplete — the scoring engine uses them as additive modifiers after the 5 MediaPipe angle calculations.

ROSA (Rapid Office Strain Assessment) is a published ergonomic scoring method from Cornell University (2011). It produces a risk score from 1-10 based on chair position, monitor height, keyboard and mouse use. This step adds 7 questions that improve the score accuracy for things the camera cannot see.

Source: all 7 inputs are official additive modifiers from the Cornell ROSA 2011 instructions. None are AI-invented.

2 POSITION IN APP FLOW

Screen	Who	Action
Consent	Employee	Existing
Pain duration / pattern	Employee	Existing
Work Pattern (+ mouse dropdown added here)	Employee	Add mouse dropdown
Your Workstation <- NEW SCREEN	Employee	Build new
Photo capture -> ROSA computed	Employee	Existing
Results screen	Employee	Existing
CPE Assessment Review	CPE Ergonomist	Existing

3 CHANGE 1 - ADD MOUSE DROPDOWN TO WORK PATTERN SCREEN

The Work Pattern screen already has dropdowns for hours at desk, break habits, and device setup. Add one more dropdown at the bottom of this screen. Same component, same style.

Property	Value
Label	Mouse type
Options	Standard mouse / Small or notebook mouse / Trackpad or no mouse
Variable name	mouseType
Component	Same dropdown as Hours at desk - reuse existing component
ROSA factor	Mouse +1 if small mouse. Mouse = 1 (no risk) if trackpad.
Source	Cornell ROSA 2011 p.17 - pinch grip on small mouse = additive +1

Device setup (Laptop / Single Screen / Dual Screen) is already on this screen. Do not change it. Dual Screen = neck twist +1 to monitor score in the ROSA engine.

4 CHANGE 2 - NEW SCREEN: YOUR WORKSTATION

Screen title: Your Workstation

Subtitle: "A few quick questions before we take the photo"

Style: same Yes/No toggle button style as the Pain duration screen already in the app.

Continue button: disabled until ALL 7 questions are answered (Q1-Q7 all have a value). Monitor distance (Q4) counts as answered when any option is selected. Back button returns to Work Pattern screen without losing data.

Q1 Can you adjust your chair height?	Options: Yes / No
Variable: chairAdjustable	ROSA: Chair +1 if No Source: Cornell ROSA 2011 p.7: non-adjustable chair = +1 additive
Q2 Is there enough leg room under your desk?	Options: Yes / No
Variable: legRoom	ROSA: Chair +1 if No Source: Cornell ROSA 2011 p.7: insufficient space under desk = +1 additive
Q3 Does your chair have lumbar support?	Options: Yes / No
Variable: lumbarSupport	ROSA: Backrest = 2 minimum if No Source: Cornell ROSA 2011 p.11: no lumbar support = backrest score 2
Q4 Monitor distance from eyes?	Options: Less than 40cm / 40-75cm / More than 75cm
Variable: monitorDistance	ROSA: Monitor +1 if more than 75cm Source: Cornell ROSA 2011 p.13: monitor must be 40-75cm (one arm's length)
Q5 Are your feet resting flat on the floor?	Options: Yes / No
Variable: feetFlat	ROSA: Chair +1 if No — seat too high or no footrest Source: Cornell ROSA 2011 p.7: insufficient foot support = +1 additive. Added because MediaPipe cannot reliably detect foot contact from landmarks (floor reference unknown to camera).
Q6 Is your monitor positioned directly in front of you (not to the side)?	Options: Yes / No
Variable: monitorFront	ROSA: Monitor +1 if No — neck twist modifier Source: Cornell ROSA 2011 p.13: neck twist >30deg = +1 additive. Added because body rotation is invisible from side profile — MediaPipe cannot detect shoulder-hip rotation from the side.

Q7 Does your chair have armrests?	Options: Yes / No
Variable: hasArmrests	ROSA: Armrest = 2 minimum if No Source: Cornell ROSA 2011 p.9: no armrests = armrest score 2. Added because MediaPipe cannot detect chair hardware — only body position consequences.

5 SCORING ENGINE - HOW TO APPLY THESE INPUTS

Apply these modifiers in the Dart score() function AFTER the 5 MediaPipe angle scores are calculated and BEFORE passing scores into lookup tables A-E. Cap rule: no individual sub-score can exceed 3 before entering the tables.

```
// --- WORKSTATION QUESTION MODIFIERS --- // Apply after angle calculations, before
lookup tables // Q1: Chair height adjustability if (chairAdjustable == false)
seatHeightScore += 1; // Q2: Leg room under desk if (legRoom == false) seatHeightScore +=
1; // Q3: Lumbar support if (lumbarSupport == false) backrestScore = max(backrestScore,
2); // Q4: Monitor distance if (monitorDistance == ">75cm") monitorScore += 1; // Q5:
Feet flat on floor // Added: MediaPipe cannot detect foot contact reliably (floor
reference unknown) if (feetFlat == false) seatHeightScore += 1; // Q6: Monitor directly
in front // Added: body rotation invisible from side profile – neck twist +1 if
(monitorFront == false) monitorScore = min(monitorScore + 1, 3); // Q7: Chair has
armrests // Added: MediaPipe cannot detect chair hardware if (hasArmrests == false)
armrestScore = max(armrestScore, 2); // Work Pattern - Dual screen (neck twist) if
(deviceSetup == "Dual Screen") monitorScore = min(monitorScore + 1, 3); // Work Pattern -
Mouse type if (mouseType == "small") mouseScore = min(mouseScore + 1, 3); if (mouseType
== "trackpad") mouseScore = 1; // Cap all sub-scores at 3 before lookup tables
seatHeightScore = min(seatHeightScore, 3); monitorScore = min(monitorScore, 3);
```

IMPORTANT: Hours at desk (from Work Pattern screen) is already captured in VAS as the duration modifier. Do NOT ask for it again on the Workstation Questions screen. Map: 0–4 hours/day = duration modifier 0. 4+ hours/day = duration modifier +1.

Data structure reference — variable names and types:

This is NOT copy-paste code. It shows the variable names, types and allowed values so you know exactly what to name things and what values to expect. Build the screen your own way using your existing patterns — just use these exact variable names so Steps 6 and 8 connect without changes.

```
// Define this object once and pass it through the assessment flow class
WorkstationInputs { bool chairAdjustable; // Q1: Can you adjust chair height? bool
legRoom; // Q2: Enough leg room under desk? bool lumbarSupport; // Q3: Chair has lumbar
support? String monitorDistance; // Q4: '<40cm' | '40-75cm' | '>75cm' bool feetFlat; //
Q5: Feet resting flat on floor? bool monitorFront; // Q6: Monitor directly in front? bool
hasArmrests; // Q7: Chair has armrests? // From Work Pattern screen (already collected):
String mouseType; // 'standard' | 'small' | 'trackpad' String deviceSetup; // 'Laptop' |
'Single Screen' | 'Dual Screen' String hoursAtDesk; // '0-4h' | '4h+' } // Hardcoded test
values for building Step 8 screen before pipeline is ready: final testInputs =
WorkstationInputs( chairAdjustable: true, legRoom: true, lumbarSupport: false,
monitorDistance: '40-75cm', feetFlat: true, monitorFront: true, hasArmrests: true,
mouseType: 'standard', deviceSetup: 'Single Screen', hoursAtDesk: '4h+', );
```

6 CPE REVIEW SCREEN - WHAT TO PASS

All 7 inputs must be stored with the assessment record and displayed on the CPE review screen as read-only fields. The CPE uses them to verify the ROSA score without re-entering any data. See Step 7 spec for full CPE screen layout.

Field	Collected on	ROSA use
Hours at desk per day	Work Pattern screen	Duration modifier
Device setup	Work Pattern screen	Neck twist modifier if Dual Screen
Mouse type	Work Pattern screen	Mouse score modifier
Chair height adjustable	Your Workstation Q1	Chair score modifier
Leg room under desk	Your Workstation Q2	Chair score modifier
Lumbar support	Your Workstation Q3	Backrest score modifier
Monitor distance	Your Workstation Q4	Monitor score modifier
Feet flat on floor	Your Workstation Q5	Chair score modifier — foot support
Monitor in front	Your Workstation Q6	Monitor score modifier — neck twist
Chair has armrests	Your Workstation Q7	Armrest score modifier

7 VERIFICATION - HOW TO CONFIRM STEP 5 IS DONE

- [] Mouse type dropdown appears on Work Pattern screen, same style as other dropdowns
- [] Your Workstation screen appears in flow between Work Pattern and photo capture
- [] Continue button stays disabled until all 7 questions answered (Q1-Q7)
- [] Run assessment with all answers = worst case (No / No / No / >75cm + Dual Screen + Small mouse)
- [] Confirm final ROSA score is higher than same photo with all best-case answers
- [] Open CPE review screen – confirm all 7 inputs visible as read-only fields
- [] All 7 values stored under the same assessment_id as the ROSA score and VAS scores

8 DEV TIME ESTIMATE

Add mouse dropdown to Work Pattern screen	30 min
Build Your Workstation screen (7 Yes/No questions + monitor distance dropdown)	3-4 hours
Wire 10 variables into score() function (Dart)	1-2 hours
Add read-only block to CPE review screen	1 hour
Total Step 5	~5-6 hours

Sources

1. Cornell ROSA 2011 Instructions – ergo.human.cornell.edu/ROSA-Instructions-2011-2012.pdf
2. Moriguchi et al. 2022, BMC Musculoskeletal Disorders – pmc.ncbi.nlm.nih.gov/articles/PMC9160176/